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Review Article

Review on the electrochemical oxidation of PFAS using Ti_4O_7 anodes

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Per- and polyfluoroalkyl substances (PFAS) are persistent organic pollutants known for their extreme environmental stability and significant biotoxicity. Magnéli-phase titanium suboxide anodes, represented by Ti_4O_7, exhibit remarkable advantages in the electrochemical oxidation of PFAS due to their high electrical conductivity, excellent electrochemical stability, and elevated oxygen evolution potential. This review systematically summarizes the recent advances in PFAS degradation using Ti_4O_7 anodes. We first analyzes the degradation mechanisms of PFAS and introduces the intrinsic advantages of Ti_4O_7 anode materials. We then highlight advanced interfacial engineering strategies to enhance the catalytic performance of Ti_4O_7, including doping, constructing composites, and structural modulation. Additionally, the article discusses the influence of water matrix components on PFAS degradation efficiency. Finally, it provides an outlook on the design of multifunctional integrated electrodes and the challenges of practical applications, offering theoretical insights and technical pathways for achieving efficient, low-energy PFAS degradation in Ti_4O_7-based electrochemical systems.

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Introduction

Per- and polyfluoroalkyl substances (PFAS) are a class of synthetic organofluoride chemicals characterized by exceptionally stable C—F bonds [1]. Among them, perfluorooctanoic acid (PFOA) and perfluorooctanesulfonate (PFOS) are the most prevalent, stemming from their broad utilization across many industries such as polymers, firefighting, and textiles [2]. Disposal of PFAS-containing products, industrial and municipal wastewater discharge, and the use of aqueous film-forming foams (AFFF), have led to widespread contamination of surface and groundwater [3]. Their environmental persistence, bioaccumulative potential, and associated acute and chronic toxicities have therefore raised serious public health and environmental concerns [4]. Consequently, there is an urgent need for effective PFAS remediation technologies.

Among emerging remediation strategies, electrochemical oxidation (EO) has attracted considerable attention for its operational simplicity, mild reaction conditions, high removal efficiency, and ease of automation [5,6]. EO primarily involve two key pathways: (i) a direct electron-transfer (DET) mechanism and (ii) an indirect oxidation route mediated by oxidizing species formed on anode such as hydroxyl radicals (•OH) [7]. A variety of anode materials such as boron-doped diamond (BDD) [8], metal oxides (e.g., SnO_2, PbO_2) [9], and carbon-based electrodes [10], have been investigated for PFAS destruction. However, each of these materials has its own drawbacks such as high cost, limited durability, and potential for toxic leaching. In this context, Magnéli phase titanium sub-oxides, particularly Ti_4O_7, have emerged as a promising alternative, which combines exceptional electrical conductivity, electrochemical stability, and cost-effective synthesis [11,12]. These attributes enable efficient PFAS degradation through following two steps: DET from PFAS to the anode followed by an indirect pathway in which PFAS-derived radicals are decomposed in solution by oxidants such as •OH generated on the anode [13].

This mini-review focuses on recent advances in the electrochemical oxidation of PFAS using Ti_4O_7-based anodes. We first outline the fundamental PFAS degradation mechanisms and advantages of Ti_4O_7 materials, then critically clarify the Ti_4O_7-based catalyst design strategy, including doping, compositing, and structural optimization. We further discuss the influence of water matrices components on PFAS degradation efficiency. Finally, we outline current challenges and future directions to facilitate the translation of Ti_4O_7-based EO systems for PFAS degradation from laboratory research to practical water treatment applications.

Mechanisms for electrochemical degradation of PFAS

The electrochemical degradation of PFAS, particularly PFOA, on Ti_4O_7-based anodes is initiated by DET from PFAS molecules to the anode surface at potentials exceeding 3.0 V *vs.* SHE, which is considered the rate-limiting step [13–15] (see Figure 1). This is followed by an indirect pathway, where PFAS-derived radicals undergo decomposition in solution through oxidants generated at the anode. Among these, •OH produced via water oxidation are widely regarded as the primary oxidizing species responsible for the indirect degradation of PFAS radicals. Once initiated, the degradation follows a stepwise "CF_2-unzipping" mechanism. The first stage involves the one-electron oxidation of PFOA, producing a perfluorinated alkyl radical. This radical quickly undergoes Kolbe decarboxylation, releasing CO_2 and forming a shorter highly active perfluoroheptyl radical ($•C_7F_{15}$). The radical reacts with •OH to form an unstable alcohol intermediate ($C_7F_{15}OH$), which then undergoes HF elimination and structural rearrangement before hydrolyzing to yield perfluoroheptanoic acid (PFHpA) with one fewer CF_2 unit. Repetition of this CF_2 cleavage cycle ultimately results in complete mineralization of the molecule into CO_2, fluoride ions (F^-), and other inorganic species.

Indirect oxidation of PFAS relies on strong oxidizing species (e.g., •OH and $•SO_4^-$) generated at the anode. These radicals diffuse into the bulk solution, where they non-selectively attack PFAS molecules, and the efficiency of this process is governed by the radical yield and their lifetime [17]. Recent studies suggest that the EO of PFAS on Ti_4O_7 can be driven by the combined action of •OH and Cl• species instead of direct anodic oxidation alone [18]. The Cl• radical is responsible for triggering the initial bond activation, rendering the initial DET step no longer rate-limiting for PFOA degradation.

Electrochemical reduction (ER) degrades PFAS via direct electron transfer and indirect pathways involving hydrated electrons (e_aq^-) and atomic hydrogen (•H). DFT calculations show that on gold electrodes, reduction follows a concerted mechanism where electron injection and C–F bond cleavage occur simultaneously, lowering the activation barrier [19]. Unsaturated C=C structures in PFAS promote defluorination, while saturated intermediates resist further breakdown. Indirect reduction by e_aq^- is rate-limited by electron transfer rather than bond scission [20]. Meanwhile, •H mediates hydrodefluorination, substituting fluorine with hydrogen. However, practical application faces challenges from competing hydrogen evolution and electrostatic repulsion of anionic PFAS at cathodes, requiring tailored electrode interfaces for efficient deep defluorination.

Electrochemical degradation of PFAS on Ti_4O_7 anodes

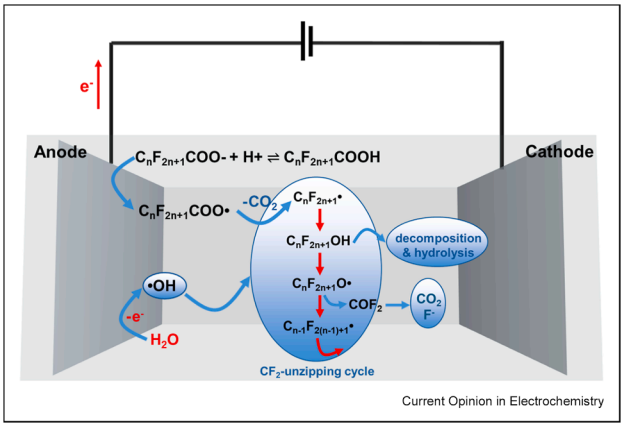
Advantages and properties of Ti_4O_7

Compared with conventional electrode materials such as BDD, Ti_4O_7-based electrodes offer significant advantages for wastewater treatment due to their low cost and unique physicochemical properties. Diamond electrodes remain prohibitively expensive (~$7000/m^2) [11], while alternatives like SnO_2 and PbO_2 suffer from short lifetimes and metal leaching risks [21,22]. In contrast, Ti_4O_7 can be synthesized from inexpensive TiO_2, enabling lab-scale fabrication at an extremely low cost (~$0.36/m^2) [6]. Moreover, the key physicochemical advantages of Ti_4O_7 include: (i) High electronic conductivity (~10^3 S/cm) from ordered oxygen vacancies forming Ti–Ti metallic bonds, ensuring efficient electron transport without carbon additives [23]; (ii) Excellent chemical and electrochemical stability, with a high oxygen evolution potential (OEP, +2.92 V vs. SHE in acidic media), enabling •OH generation while suppressing oxygen evolution [12]; (iii) Porous structure and ease of functionalization, facilitating abundant active sites and construction of reactive electrochemical membrane (REM)operations, and enabling flexible cation/anion doping to enhance carrier concentration and surface activity [24,25].

Strategies to enhance PFAS degradation on Ti_4O_7

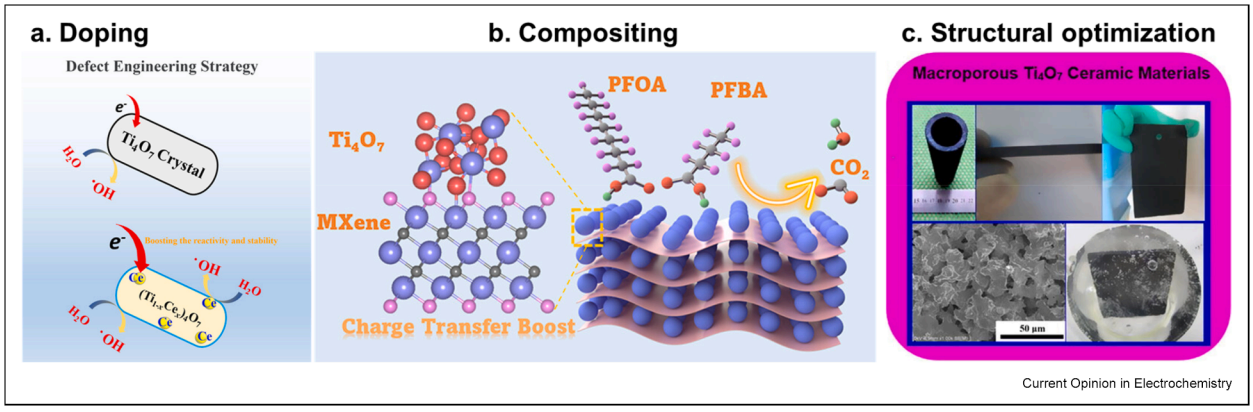
Despite the above-mentioned advantages, the intrinsic activity and interfacial charge transfer capability still limits the effectiveness of Ti_4O_7 electrode for electrooxidation of PFAS. Recently reported strategies (Figure 2), such as doping, compositing, and structural optimization, regulate the surface chemical composition, electronic structure, and interfacial interactions of Ti_4O_7 electrode, thereby effectively promoting charge transfer, exposing more active sites, and enhancing the

Figure 1



Proposed mechanisms for PFAS electrochemical oxidation via direct electron transfer and indirect •OH-mediated oxidation on a Ti_4O_7 anode. Reproduced with permission from Ref. [16].

Figure 2



Strategies for enhancing PFAS degradation on Ti_4O_7: (a) Doping with Ce^{3+}, (b) Compositing with MXene materials, and (c) Fabrication of macroporous Ti_4O_7 ceramic materials. Reproduced with permission from Refs. [25–27].

adsorption and activation of reactants. As a result, it significantly improves the activity, selectivity, and stability of Ti_4O_7-based electrodes in the electrocatalytic oxidation of PFAS pollutants (Table 1).

Doping

Doping strategies can effectively boost reactive site density, charge transfer efficiency, and structural stability [26,35,36]. Specifically, Ce^{3+} doping, owing to its larger ionic radius and different valence states with Ti, can introduce abundant surface oxygen vacancies [26]. These vacancies serve as key active sites for generating active •OH. Meanwhile, the Ce^{3+}/Ce^{4+} redox couple in Ce-doped Ti_4O_7 acts as an "electron shuttle", significantly promoting interfacial charge transfer, resulting in a 37%–129% increase in •OH yield, a 2.4-fold improvement in PFOS degradation rate, and enhanced electrode structural stability compared with pure Ti_4O_7 electrode. On the other hand, fluorine doping stabilizes oxygen vacancies by forming robust Ti–F bonds, preventing their deactivation during electrolysis [37]. This raises the oxygen defect ratio to 70.6% and boosts •OH yield by 1.38–3.20 times compared to the pristine electrode. Additionally, the hydrophobic surface introduced by fluorine facilitates PFAS adsorption and •OH desorption, enabling the electrode to maintain over 95% degradation efficiency even after 43 cycles. Most importantly, both doping strategies significantly improve electrode longevity. Ce^{3+} stabilizes oxygen vacancies, while fluorine atoms physically occupy them, shielding these sites from being passivated by reoccupied oxygen during electrolysis. This inhibits the detrimental phase transformation of conductive Ti_4O_7 into inert TiO_2, granting both doped electrodes exceptional operational stability in complex water matrices.

Compositing

Construction of Ti_4O_7-based composite electrodes can achieve superior electrocatalytic performance for pollutant degradation by enhancing charge transfer, suppressing parasitic reactions, and introducing synergistic oxidation pathways [38,39]. Two main approaches have emerged. The first focuses on improving interfacial conductivity and charge-transfer kinetics. Ti_4O_7/MXene composites leverage MXene's metallic conductivity and hydrophilicity to enhance electron transport and pollutant adsorption; density functional theory (DFT) calculations further reveal that MXene lowers •OH adsorption energy and facilitates charge transfer from PFOA, accelerating its degradation [27]. Furthermore, loading amorphous Pd clusters onto fluorinated Ti_4O_7 significantly enhances PFOA degradation by enabling a direct, non-radical anodic oxidation pathway [29]. The disordered Pd structure forms abundant Pd–O bonds with the Ti_4O_7 support, facilitating efficient electron transfer from PFOA. Surface fluorination increases hydrophobicity, promoting adsorption of PFOA and its intermediates while suppressing competitive water oxidation and •OH generation. This synergistic design achieves rapid PFOA mineralization without interference from common anions or natural organic matter, offering a robust strategy for practical PFAS treatment.

The second strategy emphasizes increasing surface reactivity and OEP to favor pollutant oxidation over water splitting. For instance, the Bi_2O_3–SnO_2/Ti_4O_7 REM combines a high-OEP SnO_2 catalytic layer with a conductive Ti_4O_7 porous substrate, enabling efficient PFOA mineralization in groundwater via both DET and •OH pathways [30]. Similarly, an N-doped graphene nanosheet-encapsulated Ti_4O_7 membrane (N-GS@Ti_4O_7) achieves an OEP exceeding 2.3 V vs. Ag/

Table 1

Summary of the recent Ti_4O_7-based anodes for electrochemical degradation of PFAS.

Electrode (anode)	PFAS	Current density	reaction time (min)	Removal efficiency (%)	Energy Consumption EE/O ($kWh \cdot m^{-3}$)	Ref.
Porous Ti_4O_7	L-PFOS	2 μM	10 mA/cm^2, 100 mM Na_2SO_4 solution	98.6% in 20 min	3.6	[28]
	B-PFOS			98.5% in 20 min	–	
Porous Ti_4O_7	PFOS	100 μM	5 mA/cm^2	93.1% in 3 h	14.2	[25]
		2 μM		>99% in 90 min	–	
$(Ti_{1-x}Ce_x)_4O_7$ (x = 3%)	PFOS	20 μM	20 mA/cm^2, $NaClO_4$ (100 mM)	>99% in 4 h	3.8	[26]
Amorphous Pd/Ti_4O_7	PFOA	120 μM	10 mA/cm^2, 50 mM Na_2SO_4 (30 mL)	86.7 ± 6.3% in 1 h	56.4	[29]
Ti_4O_7 REM	PFOA	4.14 mg/L	V anode = 3.3 V/SHE	>99.9% at J = 240 LMH	5.1	[11]
	PFOS	5 mg/L	V anode = 3.6 V/SHE	>99.9% at J = 240 LMH	6.7	
Bi_2O_3–SnO_2/Ti_4O_7 REM	PFOA	100 μM	V anode = 4.2 V/SHE, 100 mM $NaClO_4$	98.1 ± 0.4% at J = 275 LMH	45.1	[30]
N-GS/CeO_2 @Ti_4O_7 REM	GenX	2 μM	Flow-through, 20 mA cm^{-2}	~83% at a residence of 9s	3.3	[31]
Magnéli phase titanium suboxide	GenX	2 μM	Flow-through, 8 mA cm^{-2}, 20 mM Na_2SO_4 solution, 17.3 L m^{-2} h^{-1}	51.7% in 240 min	51.4	[32]
Ti_4O_7/MXene	PFOA	2.4 μM	10 mA cm^{-2}, 100 mM Na_2SO_4 solution	>99% in 3 h	8	[27]
	PFBA	4.7 μM		>97% in 3 h	–	
Ti_4O_7	PFOS	10 mg/L	10 mA cm^{-2}, 100 mM Na_2SO_4 solution	98.9% in 3 h	–	[33]
	PFOA			96% in 3 h	–	
Ti_4O_7	PFOS	2 μM	10 mA cm^{-2}, 100 mM Na_2SO_4 solution	83.5% in 1 h	–	[13]
Ti_4O_7	PFOA		50 mA cm^{-2}, coupled with ultrasound, 150 mM Na_2SO_4 solution	100% in 6 h	–	[34]
Ti_4O_7	PFOS, PFOA, FTS		40 mA cm^{-2}, 100 mM $NaClO_4$ solution	>99%	–	[14]

AgCl, comparable to or even higher than the benchmark diamond-based electrode (2.2−2.3 V vs. Ag/AgCl). In addition, it exhibits a markedly reduced Tafel slope, which collectively suppresses oxygen evolution while promoting selective, energy-efficient electro-oxidation of GenX and other PFAS in complex industrial wastewater [31].

Structural optimization

The pore structure has a decisive influence on the performance of Ti_4O_7 electrodes in degrading PFAS [39,40]. Pores >1.03 μm can effectively participate in electrochemical reactions, as they allow sufficient electrolyte penetration and provide an effective electroactive surface area (EESA) [14]. Smaller pores, due to double-layer overlap and mass transfer limitations, contribute very little to degradation. By adjusting sintering pressure and raw material particle size, the pore size distribution can be optimized. Ti_4O_7 electrodes with a hierarchical porous structure (combining large pores with suitable mesopores) exhibit higher EESA, lower charge transfer resistance, and stronger pollutant adsorption capacity, thereby significantly improving the degradation rate and mineralization efficiency of PFOA [25]. Under the REM operation mode, water flowing through the porous electrode further enhances mass transfer, making the degradation performance far superior to that of conventional batch reactors [14,40].

REM based on Ti_4O_7 effectively address the mass-transfer limitations of conventional stirred-tank electrolysis by adopting a flow-through porous anode design [11,41]. The micrometer-scale pores enable convective transport to dominate over diffusion, significantly improving pollutant delivery to active sites and accelerating oxidation kinetics [41,42]. This design provides a large electrochemically active surface area and enables ultra-efficient treatment - achieving >99.99% removal of PFOA and PFOS in a single pass with liquid residence times of less than 45 s [11]. Energy consumption is remarkably low, at only 5.1 kWh/m^3 for PFOA and 6.7 kWh/m^3 for PFOS, which is orders of magnitude lower than non-electrochemical approaches such as UV/TiO_2 or sonolysis [11,43]. Furthermore, Ti_4O_7 REM exhibit excellent stability and resilience in complex water matrices - including low-conductivity groundwater and industrial wastewater - while minimizing the formation of toxic short-chain perfluoroalkyl intermediates [11,41]. These advantages highlight their potential for deep purification and scalable deployment in real-world applications.

Other strategies

Polarity reversal (PR) of the applied current addresses two critical issues of direct current (DC) operation for EO of PFAS using Ti_4O_7 electrodes: electrode fouling and energy inefficiency [44]. Under DC, the anode undergoes phase transformation to TiO_2, reducing conductivity and defluorination efficiency. PR mitigates such ageing by periodically switching electrode roles, enabling in-situ cleaning and preserving the Ti_4O_7 surface. Optimizing the PR cycle to ∼40 min significantly enhances defluorination (e.g., PFOA: 67% vs. 53% under DC; AFFF: 40−71% vs. 29−50%), improves mass transfer, suppresses foam buildup, and maintains near-neutral pH that favors radical generation. Moreover, PR lowers energy consumption by ∼13% by exploiting the lower voltage in the initial stage of each half-cycle. Thus, PR is a promising practical strategy that simultaneously boosts PFAS degradation, prolongs electrode lifetime, and reduces energy demand, though challenges remain in parameter optimization, co-solute interference, byproduct safety, and scale-up.

Despite their benefits, the above strategies may also suffer from individual limitations. Doping risks metal leaching (Ce^{3+}) and non-selective organic adsorption (F-doping). Compositing relies on costly precious metals (e.g., Pd) and suffers from potential interfacial degradation under long-term operation. Structural optimization, particularly REM, faces pore clogging and pressure build-up in real wastewater, with laboratory energy data difficult to scale up. PR requires precise cycle optimization; its effectiveness is compromised by coexisting solutes, and the safety of halogenated byproducts remains unverified. Moreover, all strategies inadequately address competitive interference from natural organic matter and common anions (carbonate, phosphate), and show limited mineralization of short-chain PFAS (e.g., PFBA). Future designs must prioritize real-matrix validation, fouling control, byproduct toxicity, and cost-effective scalability.

Influence of water matrices

The performance of Ti_4O_7 electrodes is impacted by the water chemistry, although the rate-limiting step - DET occurs at the electrode/water interface and is primarily governed by the interfacial microenvironment. Bulk water constituents can modulate the efficiency of the indirect oxidation processes. For instance, chloride ions (Cl^-) can promote the EO of PFOA by generating chlorine radicals (Cl•), which have a low activation energy and can initiate the first degradation step [18]. The Cl• can initiate the first degradation step, and •OH was involved in the subsequent degradation of PFOA. This suggests that the PFAS degradation can occur via the synergistic effect of Cl• and •OH rather than through direct anodic oxidation in the presence of Cl^-. Notably, the Ti_4O_7 electrode significantly suppresses the formation of perchlorate compared to BDD anodes, but still leads to the generation of chlorinated organic byproducts [45]. This indicates that controlling halogenated organics remains necessary in chloride-containing water treatment using this technology.

The roles of other anions for EO of PFAS differ [46]. Nitrate (NO_3^-) at high concentrations inhibits PFAS electrooxidation by occupying active sites and scavenging •OH to inert NO_3•. Sulfate (SO_4^{2-}) slightly enhances degradation by increasing EESA and •OH production. Carbonate (CO_3^{2-}) strongly quenches •OH but generates highly selective $•CO_3^-$ via anodic oxidation, which promotes degradation through one-electron transfer. Phosphate (PO_4^{3-}) forms oxidizing phosphate radicals at low concentrations, but at higher levels it dimerizes to $P_2O_8^{4-}$, suppressing radicals and reducing efficiency. Overall, the net effect depends on EESA changes, •OH scavenging, and formation of secondary oxidants like $•CO_3^-$ or phosphate radicals.

Coexisting contaminants such as trichloroethylene obviously inhibit PFOS electrochemical degradation by competing for electron-transfer sites and •OH, while chlorinated byproducts may pose secondary pollution risks [13]. PFOS mineralization remains largely unaffected, and higher current density can partially alleviate this inhibition. Natural organic matter (NOM), such as fulvic acid (FA), also reduces the PFOA degradation efficiency by scavenging •OH and forming fouling layers on Ti_4O_7 REM surfaces, which block active sites and hinder mass transfer [47]. Despite these mechanistic insights, the present studies evaluate each co-existing substances in isolation under controlled laboratory conditions, failing to address the complex, competing interactions among multiple anions/organics and their synergistic or antagonistic effects on Ti_4O_7 performance during long-term, real-world PFAS treatment.

The PFAS degradation efficiency is also influenced by its molecular structures [48]. Long-chain perfluoroalkyl acids (PFAAs) degrade faster than short-chain ones because they have higher molecular polarizability, smaller HOMO−LUMO energy gaps, and stronger hydrophobic affinity to the anode surface, all of which facilitate DET. Besides, sulfonates (PFSAs) degrade faster than carboxylates (PFCAs) because the sulfonate group has a stronger electron-withdrawing effect, raising the HOMO energy level, and may also form specific surface interactions with the Ti_4O_7 anode.

Conclusions and outlook

Electrochemical oxidation using Ti_4O_7-based anodes represents a highly promising strategy for the effective and energy-efficient degradation of PFAS. The unique combination of high electrical conductivity, electrochemical inertness, and favorable electrocatalytic properties enables Ti_4O_7 to mineralize recalcitrant PFAS via synergistic DET and •OH-mediated pathways. Recent advances in design strategy, including Ce^{3+} and fluorine doping, MXene or N-doped graphene hybridization, hierarchical pore structuring, and polarity reversal of the applied current, have significantly enhanced degradation kinetics, selectivity, and long-term stability. Moreover, the integration of Ti_4O_7 into REM architectures has overcome traditional mass-transfer limitations, achieving complete PFAS removal in a single pass with remarkably low energy consumption.

Although significant progress has been made through doping, compositing, and structural optimization strategies, current Ti_4O_7 electrodes still face critical challenges, including limited selectivity, poor antifouling properties, and undesirable byproduct formation under real water treatment conditions. Future research should focus on developing multifunctional integrated electrodes that combine efficient adsorption, selective catalysis, in-situ monitoring, and self-cleaning within a single system. A promising approach is the design of a sandwich-structured Ti_4O_7 composite membrane electrode with an adsorption layer, catalytic layer, and conductive layer architecture. In this configuration, the adsorption layer incorporates fluorinated carbon materials or molecularly imprinted polymers to enhance PFAS-specific capture; the catalytic layer employs heterojunctions such as Ti_4O_7/g-C_3N_4 or Ti_4O_7/MOFs to enable synergistic oxidation through both radical and non-radical pathways; and the conductive layer integrates MXene or graphene networks to ensure rapid electron transport and structural stability. Such integrated electrodes are expected to achieve high selectivity, reduce energy consumption, and extend operational lifespan for PFAS degradation in complex water matrices, while minimizing the accumulation of toxic intermediates.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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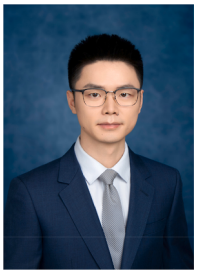
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